

		$55.0 \rho_{oil} g = (55.0 - x) \rho_{water} g \Rightarrow x = 1.5 \text{ cm}$
8	C	By conservation of energy, work done by the force $W =$ gain in kinetic energy ΔK $\frac{\Delta K'}{\Delta K} = \frac{W'}{W} = \frac{(2F)(2x)}{Fx} = 4 \Rightarrow \Delta K' = (4)(6\text{J}-2\text{J})=16 \text{ J}$ New kinetic energy $K' = 16 + 2 = 18 \text{ J}$
9	D	Area under the $F-x$ graph = work done by force. The work done increases until